Investment Demand Shock and the Business Cycle

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April 23, 2026

Background: Why investment shocks?

- Investment is highly volatile over the business cycle.
- Classic view: **supply-side** investment shocks (IST/MEI) explain a large share of macro fluctuations (e.g., Greenwood et al., 1988, Fisher, 2006, Justiniano et al., 2010).
- Puzzle: standard IST-type shocks struggle to generate the **comovement** of consumption with investment and inflation.
- Idiosyncratic productivity shocks in heterogeneous-firm models suggest a **demand-side** investment shock and can create such comovement (Nieri and Ragot, 2025).

Research question and headline result

Research question

Quantitatively, how important is an investment demand shock for the U.S. business cycle?

Main result (baseline estimation)

Investment demand shock is a competitive driver of fluctuations: it explains about **40%** of output FEVD and about **50%** of investment.

- Key qualitative contrast: ID shock generates an **inflationary** expansion, while IST shock is **deflationary** in the estimated baseline.
- Robustness: relative-price identification and a Smets–Wouters-style expansion both preserve quantitative importance of ID shock.

IST vs investment demand shock

Investment-specific technology (IST) shock

Supply-side: changes the efficiency of turning investment into capital.

$$K_{t+1} = (1-\delta)K_t + \mu_{s,t}\left[1 - S(\cdot)\right]I_t$$

Investment demand (ID) shock

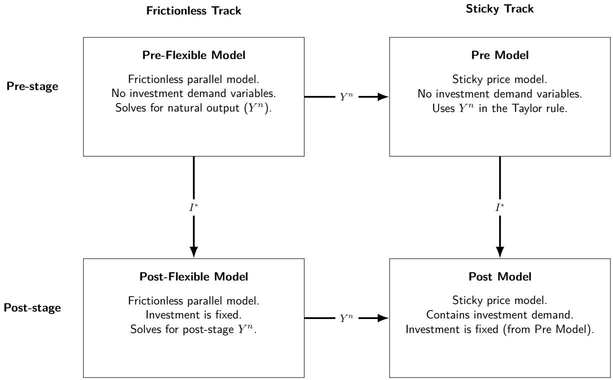
Demand-side: shifts total investment after the baseline endogenous investment decision.

$$I_t = \mu_{d,t} I_t^*$$

- Identification intuition: IST moves the (model-implied) relative price of investment; ID does not.

Two-step solution method

- **Step 1 (pre-stage):** solve the baseline DSGE without ID shock to obtain endogenous investment I_t^*.
- **Step 2 (post-stage):** reveal $\mu_{d,t}$ and set total investment $I_t = \mu_{d,t} I_t^*$; re-solve the equilibrium for other choices (consumption, labor, bonds, utilization).
- For the Taylor rule output gap, use **frictionless parallel models** to compute natural output in both stages.



Baseline DSGE model: ingredients

- Framework: estimated DSGE based on Smets and Wouters, 2007 / Herbst and Schorfheide, 2015, augmented with ID shock (two-step equilibrium).
- Nominal rigidities: Calvo price setting; sticky wage via a reduced-form contract-reset rule.
- Real frictions: investment adjustment costs; endogenous capital utilization with utilization cost.
- Policy: Taylor rule responding to inflation and the output gap; output-stabilizing fiscal policy.
- Shocks: aggregate technology, IST, monetary policy, government spending, and investment demand.

Key equations

- **Capital accumulation with IST and adjustment costs**

$$K_{t+1} = (1-\delta)K_t + \mu_{s,t}\left[1 - S\left(\frac{I_t}{I_{t-1}}\right)\right] I_t, \quad S(x) = \frac{\phi_i}{2}(x-\gamma)^2.$$

- **ID shock enters only in post-stage:** $I_t = \mu_{d,t} I_t^*$.
- **Endogenous capital utilization:** $K_t^s = u_t K_{t-1}$.
- **Government spending:** $\hat{g}_t = -\phi_g \hat{y}_t + \hat{\mu}_{g,t}$
- **Sticky wages (baseline):** partial adjustment toward MRS

$$w_t = \left(\frac{L_t^{\psi}}{C_t^{-\sigma}}\right)^{\theta_w} w^{1-\theta_w}.$$

Estimation: data (baseline)

- Six observables (quarterly):
  - per-capita real output growth
  - CPI inflation (annualized)
  - effective federal funds rate
  - (log) hours worked
  - per-capita real investment growth
  - per-capita real government spending growth
- Measurement error used to mitigate stochastic singularity.

Summary: What changed vs the previous version of the slides?

- This version uses two different frictionless parallel models to solve natural output separately in pre-stage and post-stage.
- Stochastic growth trend is changed to deterministic growth trend.
- Adds investment adjustment cost, and endogenous capital utilization with utilization cost.
- Data set includes hours worked, investment growth and government spending growth.
- Robustness is formalized in two directions: (i) add **relative price of investment** to discipline IST; (ii) expand toward a **Smets–Wouters-style** model with extra shocks and extra observables.

Model fit: moments (baseline)

- The model matches most means/variances reasonably.
- Main tension: implied variance of hours is too high, suggesting missing frictions (e.g., habit) would dampen labor volatility.

Table: Model-implied and data moments (mean and variance) for the baseline observables (output, inflation, interest rate, hours, investment, and government spending).

	Mean		Variance	
Observable	Model	Data	Model	Data
Output	0.33	0.56	0.67	0.34
Inflation	4.05	3.08	5.09	2.16
Interest rate	7.15	6.05	8.33	5.01
Hours	0.96	0.00	31.86	6.37
Investment	0.33	0.57	3.87	2.53
Gov. spending	0.33	0.31	0.82	0.85

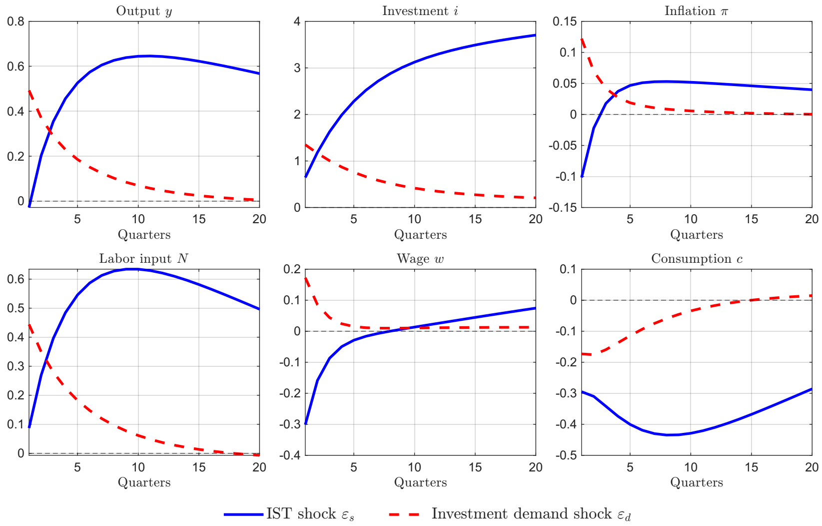
Forecast error variance decomposition (baseline)

- IST + ID together explain about **65%** of output FEVD, **45%** of inflation, and **90%** of hours.
- ID shock alone: about **40%** of output FEVD and about **50%** of investment.

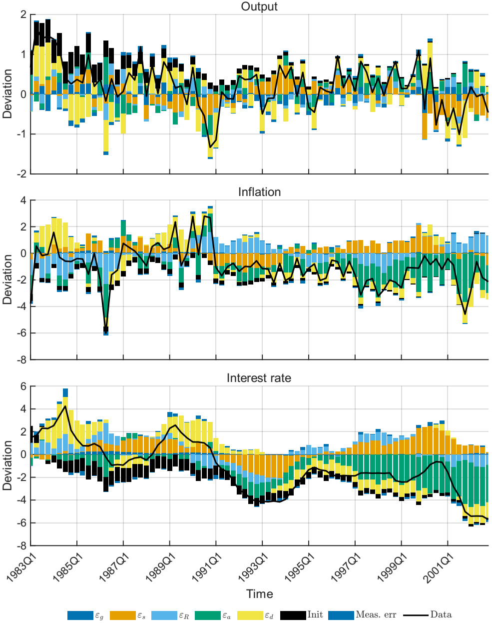
Table: Unconditional FEVD (%) for the baseline estimation, by government spending (ε_g), IST (ε_s), monetary policy (ε_R), technology (ε_a), and investment demand (ε_d) shocks.

Observable	ε_g	ε_s	ε_R	ε_a	ε_d
Output	4.63	15.55	11.00	27.12	41.69
Inflation	0.32	39.26	12.87	40.14	7.41
Interest rate	0.37	55.06	5.48	31.22	7.87
Hours	4.34	90.19	0.08	3.56	1.83
Investment	0.11	36.23	0.15	12.04	51.48
Gov. spending	99.87	0.02	0.01	0.04	0.06

Impulse responses: inflationary ID vs deflationary IST



Historical decomposition (baseline)



Robustness I: relative price disciplines IST

- Add relative price of investment as an observable and interpret detrended relative price as inverse IST.
- Result: the inferred volatility/importance of IST falls; ID shock becomes more important in explaining macro fluctuations.

Table: Unconditional FEVD (%) for the model estimated with the relative price of investment as an additional observable, by structural shocks.

Observable	ε_g	ε_{ist}	ε_R	ε_a	ε_{id}
Output	5.72	0.59	17.21	17.86	58.61
Inflation	0.78	2.84	13.15	73.26	9.97
Interest rate	1.10	4.06	7.02	74.88	12.94
Hours	7.74	8.29	2.25	38.17	43.55
Investment	0.21	1.73	0.59	9.61	87.86
Gov. spending	99.44	0.00	0.10	0.11	0.35
Relative price	0.00	100.00	0.00	0.00	0.00

Robustness II: Smets–Wouters-style expansion

- Add key SW mechanisms: habit formation, monopolistic labor market structure, and additional shocks (preference + markup shocks).
- Add extra observables: consumption and wage growth.
- ID remains quantitatively important: about **25%** of output FEVD and **78%** of investment.

Table: Unconditional FEVD (%) for the Smets–Wouters-style robustness model, by government spending, IST, monetary policy, technology, preference, price-markup, wage-markup, and investment demand shocks.

Observable	ε_g	ε_{ist}	ε_R	ε_a	ε_b	ε_p	ε_w	ε_{id}
Output	5.80	0.29	5.49	39.32	14.38	4.07	4.88	25.77
Inflation	0.04	0.24	0.45	14.95	0.13	56.30	27.06	0.83
Interest rate	0.21	0.93	30.33	24.94	3.57	11.91	25.21	2.89
Hours	1.79	1.66	5.15	46.11	5.85	0.91	29.13	9.40
Investment	0.20	2.61	1.32	10.97	1.30	0.38	5.06	78.15
Wage	0.00	0.01	0.10	16.82	0.17	16.09	66.77	0.03
Consumption	0.26	0.39	5.31	50.58	36.61	1.72	4.08	1.05
Gov. spending	99.91	0.00	0.01	0.04	0.01	0.00	0.00	0.02
Relative price	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00

Takeaways

- Estimated evidence supports an investment-demand-side driver of fluctuations in addition to IST.
- Baseline: ID shock delivers sizable FEVD shares and an inflationary expansion (vs IST deflationary).
- Robustness checks (relative price + SW-type model) preserve the main conclusion.

References I

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If we drop investment adjustment costs

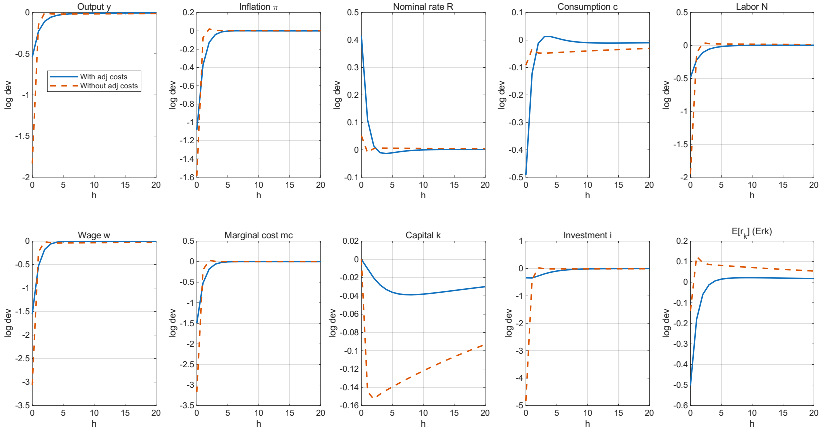

Figure: IRF: 1 s.d. monetary shock in the model with and without investment adjustment costs

If we don't use government spending data

Table: Unconditional Variance decomposition (%).

Observable	ε_g	ε_{ist}	ε_R	ε_a	ε_{id}
Output	51.00	10.06	11.40	19.72	7.82
Inflation	15.19	37.49	16.06	30.65	0.61
Interest rate	20.64	47.73	5.36	25.92	0.35
Hours	85.73	11.27	0.43	2.25	0.32
Investment	0.24	61.13	0.90	14.32	23.40